

THE CHAIN RULE

$$y' = f'(g(x)) \cdot g'(x)$$

Note: The composite function $(f \circ g)$ is defined by $(f \circ g)(x) = f(g(x))$

$$f \circ g \neq g \circ f$$

Example 1: Suppose $f(x) = \sqrt{x}$ and $g(x) = x - 2$

a) Express $(f \circ g)(x)$

$$(f \circ g)(x) = \sqrt{x-2}$$

c) Express $(g \circ f)(x)$

$$(g \circ f)(x) = \sqrt{x} - 2$$

Not commutative

b) Evaluate $f(g(4))$

$$f(g(4)) = \sqrt{2}$$

d) Evaluate $g(f(4))$

$$g(f(4)) = 0$$

Example 2: Differentiate the function $h(x) = (2x^3 + 5x)^{35}$

$$h'(x) = 35(2x^3 + 5x)^{34} \cdot (6x^2 + 5)$$

Note:

If we let $f(u) = u^{35}$ and we let $u = g(x) = 2x^3 + 5x$ then $h(x) = f(g(x))$.

Since we know how to differentiate $f(u)$ and $g(x)$ individually, the **chain rule** allows us to differentiate $y = f(g(x))$.

It turns out that the derivative of the composite function, $h(x) = f(g(x))$ is the product of the derivatives of $f(u)$ and $g(x)$.

That is, $h'(x) = f'(u)g'(x)$.

$$h'(x) = 35u^{34} \cdot (6x^2 + 5)$$

$$h'(x) = 35(2x^3 + 5x)^{34} \cdot (6x^2 + 5)$$

Recall:

$\frac{dy}{du}$ is the rate of change of y with respect to u .

$\frac{du}{dx}$ is the rate of change of u with respect to x .

$\frac{dy}{dx}$ is the rate of change of y with respect to x .

If $g(x)$ is differentiable at x and $f(x)$ is differentiable at $g(x)$, then the composite function, $h(x) = f(g(x))$ or $h(x) = (f \circ g)(x)$ is differentiable at x and $h'(x)$ is given by the product $h'(x) = f'(g(x))g'(x)$.

In Leibniz notation, If $y=f(u)$ and $u=g(x)$, are both differentiable functions, then

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

Examples:

1. If $y = 2u^2 + 6u$, $u = 2 + v^3$, and $v = \sqrt{x}$ then evaluate $\frac{dy}{dx}$ at $x = 4$.

$$\left(\frac{dy}{dx} \right) = \left(\frac{dy}{du} \right) \cdot \left(\frac{du}{dv} \right) \cdot \left(\frac{dv}{dx} \right)$$

when $x = 4$,

$$v = 2,$$

$$u = 10$$

$$\frac{dy}{dx} = (4u + 6) \cdot (3v^2) \cdot \frac{1}{2\sqrt{x}}$$

$$\therefore \left. \frac{dy}{dx} \right|_{x=4} = \frac{(46)(12)}{4} = 138$$

2. Determine the derivative of $f(x) = \sqrt{\frac{x-3}{3-5x}}$ let $u = g(x) = \frac{x-3}{3-5x}$

$$* f'(x) = h'(u) \cdot g'(x)$$

$$\therefore h(u) = \sqrt{u}$$

$$= \frac{1}{2\sqrt{u}} \cdot \left[\frac{(1)(3-5x) + 5(x-3)}{(3-5x)^2} \right]$$

$$= \frac{-12}{2\sqrt{\frac{x-3}{3-5x}} \cdot (3-5x)^2}$$

$$= \frac{-6}{\sqrt{x-3} \cdot \sqrt{(3-5x)^3}}$$

$$= \frac{-6}{\sqrt{(x-3)(3-5x)^3}}$$

3. Differentiate:

a) $f(x) = (4x^2 + 3x)^5$

$$f'(x) = 5(4x^2 + 3x)^4 \cdot (8x + 3)$$

b) $f(x) = (5x^2 - x)^\pi$

$$f'(x) = \pi(5x^2 - x)^{\pi-1} \cdot (10x - 1)$$

c) $y = \frac{x}{1-4x^2}$

$$y' = \frac{1-4x^2 - x(-8x)}{(1-4x^2)^2}$$

$$y' = \frac{1+4x^2}{(1-4x^2)^2}$$

d) $f(t) = (2t^3 + 6t - 5)^5$

$$f'(t) = 5(2t^3 + 6t - 5)^4 (6t^2 + 6)$$

$$f'(t) = 30(2t^3 + 6t - 5)^4 (t^2 + 1)$$

e) $y = (3x^2 - 5)\sqrt{2x^2 + 9x}$

$$\frac{dy}{dx} = 6x\sqrt{2x^2 + 9x} + \frac{(3x^2 - 5) \cdot (4x + 9)}{2\sqrt{2x^2 + 9x}}$$

$$\frac{dy}{dx} = \frac{12x(2x^2 + 9x) + (3x^2 - 5)(4x + 9)}{2\sqrt{2x^2 + 9x}}$$

$$\frac{dy}{dx} = \frac{24x^3 + 108x^2 + 12x^3 + 27x^2 - 20x - 45}{2\sqrt{2x^2 + 9x}}$$

$$\frac{dy}{dx} = \frac{36x^3 + 135x^2 - 20x - 45}{2\sqrt{2x^2 + 9x}}$$

$$\begin{aligned} & (2x^2 + 9x)^{\frac{1}{2}} \rightarrow \frac{1}{2}(2x^2 + 9x)^{-\frac{1}{2}} \cdot (4x + 9) \\ & = \frac{4x + 9}{2(2x^2 + 9x)^{\frac{1}{2}}} = \frac{4x + 9}{2\sqrt{2x^2 + 9x}} \end{aligned}$$

f) $m(t) = \sqrt[3]{1+t+t^2}$

$$m'(t) = \frac{1}{3 \cdot \sqrt[3]{(1+t+t^2)^2}} \cdot (1+2t)$$

$$= \frac{2t + 1}{3 \cdot \sqrt[3]{(1+t+t^2)^2}}$$

4. a) If $h(x) = \frac{(f(x))^2}{g(x)}$, determine $h'(x)$.

$$h'(x) = \frac{2f(x) \cdot f'(x) \cdot g(x) - (f(x))^2 \cdot g'(x)}{[g(x)]^2}$$

b) Given $f(1)=2$, $f'(1)=-3$, $g(1)=1$ and $g'(1)=4$, determine $h'(1)$.

$$h'(1) = \frac{2f(1) \cdot f'(1) \cdot g(1) - (f(1))^2 \cdot g'(1)}{[g(1)]^2}$$

$$h'(1) = \frac{2(2)(-3)(1) - (2)^2 \cdot (4)}{(1)^2}$$

$$h'(1) = -12 - 16$$

$$h'(1) = -28$$